

Neural Networks

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Course Overview

This course provides a deep dive into neural networks, starting from foundational concepts and progressing to advanced topics such as deep learning architectures, optimization techniques, and real-world applications. The course includes theoretical lectures and projects to reinforce learning.

Prerequisites

- Basic knowledge of linear algebra, calculus, and probability.
- Familiarity with Python programming.
- Introductory understanding of machine learning concepts.

Course Outline

Module	Lecture	Topics
1: Neural Network Foundations and Learning Paradigms	Lecture 1	Introduction to Neural Networks: History, biological motivation, modern AI applications.
	Lecture 2	Supervised Learning Basics: Perceptron model, loss functions, gradient descent training.
	Lecture 3	Multi-Layer Perceptrons (MLPs): Forward/backpropagation, classification vs regression tasks.
	Lecture 4	Training Challenges and Regularization: Overfitting, bias-variance tradeoff, dropout, early stopping.
2: Deep Architectures for Supervised and Unsupervised Learning	Lecture 5	Convolutional Neural Networks (CNNs): Feature extraction, supervised image recognition tasks.
	Lecture 6	Sequence Models (RNNs, LSTMs, GRUs): Temporal learning for speech, text, and sensor data.
	Lecture 7	Transformers and Attention: Self-attention, encoder-decoder models, BERT and GPT applications.
	Lecture 8	Unsupervised and Generative Learning: Autoencoders, VAEs, GANs, representation learning.
3: Training Process, Explainability, and Hybrid Intelligent Systems	Lecture 9	Advanced Optimization and Training Pipeline: Adam, RMSprop, batch normalization, learning schedules.
	Lecture 10	Transfer Learning and Semi-Supervised Learning: Fine-tuning, domain adaptation, limited-label training.
	Lecture 11	Explainable AI (XAI): Interpretability methods (LIME, SHAP, Grad-CAM), trust and transparency.
	Lecture 12	Fuzzy Systems and Neuro-Fuzzy Networks: Fuzzy logic, rule-based reasoning, ANFIS integration.
4: Project	Lecture 13	End-to-End AI Solution Development: Data preparation, supervised/unsupervised training, explainability, deployment.

Tentative Assessment and Grading

- **Paper Review (10%):** Weekly paper review assignments to reinforce lecture topics.
- **Quizzes (15%):** Short quizzes to test theoretical understanding.
- **Project (15%):** Evaluation based on project design, implementation, and presentation.
- **Midterm Exam (25%):** Covers foundational and intermediate topics.
- **Final Exam (35%):** Covers foundational and intermediate topics.

Recommended Resources

- **Textbooks:**
 - "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
 - "Neural Networks and Deep Learning" by Michael Nielsen.
 - "Neural Networks and Deep Learning" by Charu C. Aggarwal.
- **Online Courses:**
 - Deep Learning Specialization by Andrew Ng (Coursera).
 - Fast.ai Practical Deep Learning for Coders.
- **Frameworks and Tools:**
 - TensorFlow, PyTorch, Keras.
 - Jupyter Notebook, Google Colab.

Conclusion

This course equips students with the knowledge and skills to design, implement, and deploy neural networks for a wide range of applications. By combining theoretical foundations with hands-on practice, students will gain the expertise needed to tackle real-world challenges using advanced neural network techniques.